

Adaptive Class-Balanced Loss Based on Re-Weighting

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Abstract—As real-world data grows fast, the problem of data imbalance has become more prominent. Thus the long-tail problem in deep learning has received lots of attention recently. One of the solutions is to apply a class rebalancing strategy, such as directly using the inverse of the class sample size for reweighting. In past studies, the setting of weights only relates to the number of class samples. Only relying on the information of the number of class samples to determine the size of the weight is very crude in the sensitive method of re-weighting. In this paper, we implement adaptive re-weighting for three essential attributes of the dataset considering several factors: the number of classes, the number of samples, and the degree of class imbalance. We conducted experiments on the commonly used sample imbalance problem solution and proposed a new sample reweighting method. Specifically, a novel re-weighting idea is proposed to optimize Class-Balanced Loss Based on an Effective Number of Samples. Experiments show that the method is superior in re-weighting imbalanced datasets on deep neural networks. We hope our work will stimulate a rethinking of the number-of-samples-based convention in re-weighting.

Index Terms—Deep learning, Long-tail, Data imbalance, Re-weighting

I. INTRODUCTION

The research of computer vision accompanied by deep convolutional neural networks [1]–[4], and the development of large-scale image datasets [5]–[7] has been a great success. The number of samples per class in the training set is often manually balanced in the commonly used datasets. In the real world, the distribution of datasets is often extremely unbalanced [8], [9]: Usually, a few dominant classes (i.e., head classes) constitute the major of the samples in the entire data set, while most of the disadvantaged classes (i.e., tail classes) constitute merely a relatively small number of samples.

Class imbalance is widespread in the real world, and sometimes it is important to classify a rare class correctly. For example, lesions barely occur in hospital visits, but the cost is enormous once this disease is misdiagnosed as a common

disease. Similarly, in computer vision fields such as image classification, the correct classification of rare classes is as important as the correct classification of other classes. As for the detection problem, the distribution of background pictures and foreground pictures is also highly imbalanced, but the correctness of a few foreground pictures is essential.

Under the condition of imbalanced classes, there is an objective fact that cannot be ignored. In some classes, due to various factors, the sample is always in small number. This is an objective fact that cannot be changed in the data. The model comes from training a large amount of labeled data in the supervised learning method. When certain data types are objectively scarce, its recognition effect will inevitably be affected. However, the lack of data cannot be compensated no matter from which point of view the optimization is carried out. In the current work, the harm caused by unbalanced data can only be mitigated as much as possible but cannot be solved entirely.

Recent research has shown that when training models using imbalanced data, the head classes with major instances could dominate the training; thus, the classification models would perform poorly on the tail classes [10]–[13]. Many recent studies focus on mitigating the challenges posed by such kind of long-tail data [14]–[19]. In [14], the authors take re-sampling approaches to adjust the sample size by either under-sampling the head class or oversampling the tail class, or both methods jointly. In [15], the authors provide a re-weighting mechanism to adjust the loss function by assigning different costs to different numbers of head and tail classes. Classification performance can also be improved by transferring the knowledge of the head class to the tail class. More recently, the network's performance is improved by training the representation separately from the classification. In general, these existing approaches aim to design new loss and even more complex network models using appropriate sampling strategies, which can be helpful for long-tail recognition.

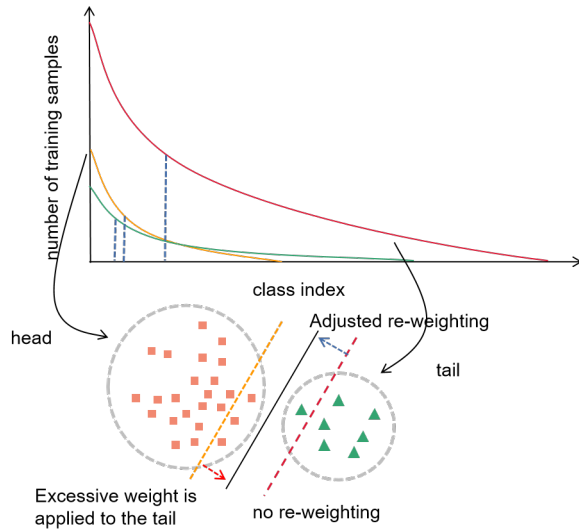


Fig. 1. In the long-tail data set, the major class is called the head class and the minor class is called the tail class. Obviously, training the model without any further process will incline the model toward the dominant class (red dashed line). Re-weighting by inverse class frequency, etc., may lead to over-enhancement of the tail class and over-suppression of the head class (orange dashed line).

In existing re-weighting methods, the coefficients for weighting are usually determined based on the number of class samples, which is concise and effective. A weighting balance is achieved by applying a smaller weight to the head class and a larger weight to the tail class. This paper investigates whether previous weighting approaches are still effective when the number of classes rises and the degree of imbalance changes.

It is well known that in the classification problem, classes and sample size are two critical factors in the neural network. While more classes lead to higher training difficulty in the neural network, more samples can improve accuracy. Our study found three inherent properties of the dataset: the number of classes, the number of samples, and the data imbalance. These three factors have a significant impact on the accuracy of classification. Past researches focus mainly on the effect of sample size rather than three aspects altogether.

The re-weighting method achieves the loss of balance through a designed weighting function or setting a weighting coefficient through prior knowledge. We use the inherent attributes of the data set as previous information instead of applying additional prior knowledge to the adaptive model for data training.

Overall, it is found that in a practical imbalanced task, re-weighting requires a reconsideration of the factors influencing the determination of the weighting coefficients. Our contributions are as follows:

- We have investigated the application scenarios of common sample imbalance solutions in a more systematic way
- We are the first to include data imbalance with the number of classes in the re-weighting calculation.
- We investigated the effect of sample size and the number

of classes on imbalance learning.

- Use our ideas to optimize the Class-Balanced loss [20] and achieve good results.

II. RELATED WORK

With the development of deep learning, CNN has been widely applied to different area [36]–[38], but the problem of data imbalance is prevalent in the real world. Therefore, long-tail identification has received more and more attention in recent years [21]–[24]. Research methods on the long-tail problem can be summarised as follows:

Sample re-balancing strategy: Re-sampling method is one of the most common methods to achieve a more balanced data distribution by re-sampling the original data set. Re-sampling includes oversampling of tail classes and under-sampling of head classes. The under-sampling method [10], [12] samples the head classes by randomly discarding some of them, which may lead to the problem of information loss by head classes. Oversampling [25]–[27] method strengthens the tail class by replicating samples, but it may lead to overfitting in the tail class. SMOTE [28] proposes to synthesize the tail class data to achieve the enhancement of the tail class data, thus reducing the risk of overfitting the tail class data.

The resampling method adjusts the data imbalance from the aspect of the total number of samples. No matter how the sampling is performed, the size of the original image feature space is constant. It is impossible to introduce more class information, and under-sampling will undermine the feature space of the original head class.

Cost-sensitive strategy: The idea of weighting the training samples has been well studied before. The concept comes from importance sampling in statistics [29] where different weights are assigned to different samples to fit the data distribution. In the case of the data imbalance dataset, the intuitive idea is to use inverse class frequency weighting [16], [17]. Furthermore, there exists smoothing weighting by inverting the square root of class frequencies [30], [31]. Recently [20] a Re-weighting method is proposed to recalculate the number of effective samples instead of class frequencies. The above weighting approaches all decide the size of the assigned weights by the number of samples. In the classification task, the class number also determines the difficulty of the study. In the long-tailed data set, the imbalance has an enormous impact on the classification difficulty.

In previous work, there has been no research on these two aspects. In this paper, we incorporate the number of classes and imbalance into the re-weighting consideration. The re-weighting strategy is examined from a new perspective.

Two-stage approach: Some recent work separates feature learning from classifier learning [22], [23] and trains both respectively. Splitting imbalance learning into two stages, feature learning and classifier learning, by different sampling methods. Related experiments prove that in long-tail recognition, it is advantageous to readjust the classifier's decision boundary during the representation learning period. It can bring good

long-tail learning results. Its complexity and non-end-to-end training mode are often difficult to use in real-world scenarios.

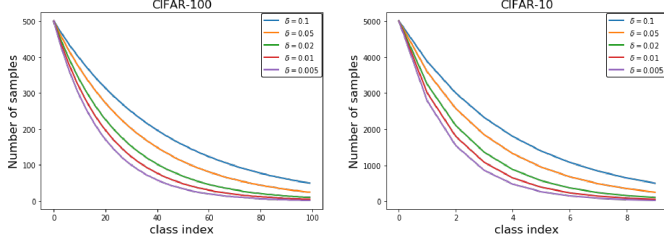


Fig. 2. Example distribution of long-tailed datasets under different.

III. RE-WEIGHTING METHOD

A. Forms of imbalance

There are various forms of class imbalance in deep learning. As for multiple classifications, it can be a shortage of over-sampling of individual classes. Another possibility is that each class has a different number of samples. In this paper, we focus mainly on the long-tail form of class imbalance.

In a long-tail imbalance, a few classes have dominant samples while most classes have small sample sizes. This imbalanced dataset is defined by the parameter δ as follows:

$$N_i = N_{max}(\delta^{\frac{i}{C-1}}), i \in \{0, \dots, C-1\} \quad (1)$$

where N_i is the number of samples in class i . C is the total number of classes. N_{max} is the maximum number of samples. We follow this paradigm to define the long-tail dataset

B. Weighting Method

In machine learning, the basic assumption of the classifier approach is based on the fact that the number of samples in different classes of the data set under study is approximately balanced. That is, the prior probabilities of each class are approximate. Therefore, according to this basic assumption, the cost of misclassification is approximately the same for all classes [33].

Re-weighting means applying different weights to each class, thus changing the cost of class misclassification. Applying higher weights to the classes that require attention makes the classifier more sensitive to the classes of interest.

Usually, we decide how to apply the corresponding weights based on the number of samples in the class. The weights can be easily obtained by the inverse class frequency method. The weights are larger for a small number of classes and smaller for most classes.

To explain better, we define D_1 and D_2 to represent two different data sets respectively. Here we assume that the two different data sets have the same imbalance degree of 0.01

$$\delta = D_1^{tail} / D_1^{head} = D_2^{tail} / D_2^{head} = 0.01 \quad (2)$$

$$\omega_i^{head} = \varphi(D_i^{head}), i \in \{0, \dots, n-1\} \quad (3)$$

$$\omega_i^{tail} = \varphi(D_i^{tail}), i \in \{0, \dots, n-1\} \quad (4)$$

$$\omega_i^{head} / \omega_i^{tail} \rightarrow \delta \quad (5)$$

$\omega_i^{head}, \omega_i^{tail}$ denote the weighting coefficients of the head and tail data in the i -th data set, respectively. φ is the linear transformation for the number of classes.

In long-tail datasets, the number of samples in the head class is often much larger than the number of samples in the tail class, which means the discrepancy in terms of sample size for different classes could be enormous. In this case, the previous re-weighting method may lead to overfitting of the tail class data or underfitting of the head class data, i.e., when $D_1^{head} \gg D_2^{head}, D_1^{tail} > D_2^{tail}$.

$$\omega_2^{head} \rightarrow \delta \times \omega_2^{tail} \quad (6)$$

If the number of D_2^{head} itself is not sufficient in this case, the same weighting measure as D_1^{head} is taken, which will result in inadequate feature learning of D_2^{head} . Also D_2^{tail} data are relatively small, and giving too much weight will lead to its severe overfitting.

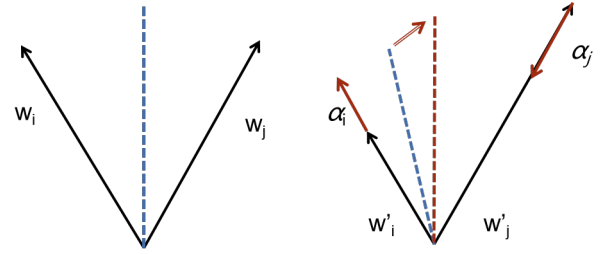


Fig. 3. Initially or when the class are uniform, the weight norms of the i -th and j -th class are as shown in the left figure, and the dashed line is the decision boundary. When the classes are unbalanced, the weight norm of the minor class i will decrease, and the major class j will have a larger volume in the feature space. We adjust the size of the feature space of the class through the weighted coefficients α_i and α_j .

As discussed in [35], the weight norm of the head class is greater than the weight norm of the tail class. As shown in Figure 3, the head class often has a large distinguishing volume in the feature space. The re-weighting in the softmax loss form is as follows:

$$L = -\alpha_i \frac{1}{N} \sum_{i=1}^N \log \frac{e^{W_{y_i} x_i}}{\sum_{j=1}^n e^{W_j x_i}} \quad (7)$$

Further:

$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{\alpha_i W_{y_i} x_i}}{\sum_{j=1}^n e^{\alpha_j W_j x_i}} \quad (8)$$

Where y_i is the corresponding class label and W is the weight for the last layer of the network which acts as a classifier. For simplicity, the bias in the formula is all 0. We can transform the output logits as $\alpha_i W_j x_i = \alpha_i |W_j| |x_i|$. The re-weighting parameter α_i can be regarded as an adjustment to the weight norm. When the network weight is initialized, the size of the weight vector norm tends to be the same. Since the imbalance of the classes causes the weight norm of the tail class to be compressed, prior weighting information needs to be set at the beginning of the network training. This weighted information is often carefully designed by humans.

In [20], the size of the weighting coefficient is re-adjusted by calculating the number of effective samples in each class. The growth rate of the number of effective samples E_n is controlled by the hyperparameter β , which enables the calculation of effective samples under different data sets. The effective sample size is defined as follows:

$$E_n = (1 - \beta^n)/(1 - \beta) \quad (9)$$

$$\beta = N - 1/N \quad (10)$$

The introduction of additional hyperparameters allows us to tune the network more carefully. They require manual human tuning, which can be time-consuming and complicated, as shown in Table 1. The tuning by hyperparameters β is too artificial. Several adjustments are needed to ascertain the most appropriate β value. We show that the growth rate of the effective sample size E_n can be determined automatically by the number of samples, the number of classes, and the degree of imbalance without any manual parameter adjustment.

C. Factors in the dataset

Deep learning is a subject about data. Being able to get more information from the data set and use it usually leads to better results. As discussed in the previous section, under the condition of unbalanced class data, we often need to design a weighting coefficient manually.

A Classification dataset often has several salient features: the number of classes, the number of samples, and the data set's sample distribution (degree of class imbalance). At the same time, the scales between these features are not the same. Existing solutions to the problem of sample imbalance usually focus on the issue of sample size. The size of the weighting coefficient is determined by the number of samples in each class. The hyperparameter β in Table 1 clearly shows that the size of the weighting coefficient does not only depend on a single dataset factor.

As shown in Table 1 and Table 2, in the classification task, we can easily observe the impact of the number of classes, the number of samples, and the degree of imbalance on the accuracy. So we come up with a very natural idea, which is different from the previous work. We consider using the salient features of the dataset itself to adjust the size of the weighting coefficient adaptively.

D. Adaptive class-balanced loss

We improve the class-balanced loss to address the drawback mentioned before. The corresponding formula is designed to calculate β by the three factors of sample size, class size and degree of class imbalance. We define it as follows:

$$\beta = F(f(u, v, b)) \quad (11)$$

where $f(u, v, b)$, is defined as follows:

$$f(u, v, b) = \frac{u}{v\sqrt{b}} \quad (12)$$

$$u = \log \sum N_i \quad (13)$$

$$v = \log C \quad (14)$$

$$b = -\frac{1}{C} \left(\sum_i^C \log_{10} \frac{N_i}{N_{max}} \right) \quad (15)$$

We control the number of samples, the number of classes and the imbalance degree on β by u, v, b respectively, To dynamically control the number of effective samples E_n . An obvious problem is that the scales of these three elements are inconsistent. We design a mapping F so that the distribution of the results obtained is consistent with the original β .

We do not consider the specific number of samples and the particular number of classes directly. Instead, use u, v to represent the scale of the number of samples and the number of classes. Thus, we can evaluate the scale of the average sample per class in the data set by $\frac{u}{v}$. Compared to the direct division by specific quantities, the quantity scale slows down the effect of the difference in quantities before the major and minor classes on the mean number of classes in the data set. Obtaining the data scale through the logarithmic method can well scale large magnitudes so that larger numbers can drop rapidly, and the scaling rate of smaller numbers is relatively flat. The logarithmic form effectively alleviates the problems caused by the mismatch of numerical scales.

To measure the degree of class imbalance of the dataset, we introduced [34] to use entropy to measure the degree of class imbalance of the dataset. Taking into account the huge difference between the major class and the minor class, $\log_{10}$ is used to calculate the entropy. When the imbalance of the data set increases significantly, its entropy also gradually increases. When $\delta \rightarrow 1$ is $N_i \rightarrow N_{max}$, $E_n \rightarrow n$. When $\delta \rightarrow 0$ is $N_i \rightarrow 1$, $E_n \rightarrow 1$. Through the change of entropy, we can use it as a fine-tuning of the scale of the class sample.

To achieve the purpose of the number of samples, the number of classes, and the degree of imbalance, the three together act on the weighting coefficient. We use the Tanh function F for further transformation to fit the numerical distribution of the original β parameters. Unlike the original CB loss, which requires manual setting of hyperparameters β , our method can determine the appropriate weighting parameters by obtaining the information of the data set. To achieve the purpose of adaptively determining the weighting coefficient.

The following is the implementation of the adaptive class-balanced loss in each loss function.

$$ACB_{softmax}(x, y) = -\frac{1 - \beta}{1 - \beta^{n_i}} \log \left(\frac{\exp(x_{y_i})}{\sum_{j=1}^C \exp(x_j)} \right) \quad (16)$$

$$ACB_{sigmoid}(x, y) = -\frac{1 - \beta}{1 - \beta^{n_i}} \sum_{i=1}^C \log \left(\frac{1}{1 + \exp(-x_i)} \right) \quad (17)$$

$$ACB_{focal}(x, y) = -\frac{1 - \beta}{1 - \beta^{n_i}} \sum_{i=1}^C (1 - p_i)^\gamma \log(p_i) \quad (18)$$

Where β is expressed as equation (11). x is the predicted output from the model, C is the total number of classes. Suppose class y has n_i training samples.

IV. ANALYSIS OF COMMON METHODS

A. Datasets and models

In this Experiment, we used three benchmark datasets: CIFAR-10, CIFAR100, and ImageNet-Tiny. Both of them provide splits on training and test sets. We choose ResNet18 as the experimental model.

Tiny-ImageNet is an image classification dataset provided by Stanford University. They contain 64×64 color images of 200 classes of natural objects. The standard Tiny-ImageNet dataset has 500 training and 50 testing samples per class.

B. Evaluation of CIFAR dataset

Table 3 shows the impact of class imbalance on classification performance and a comparison of methods to address the imbalance. It shows F1-score results for different degrees of imbalance on CIFAR-10 and CIFAR-100. Regarding the effect of class sample imbalance on classification performance, from the results, we observe the following points. First, the performance deterioration due to class imbalance is quite severe. As expected, the increasing ratio of samples between the majority and minority classes and the number of minority classes negatively affected the performance of the resulting classifier. In addition, by comparing the test results of CIFAR-10 and CIFAR-100, we found that the more complex tasks, the more obvious the imbalance effect.

Regarding the performance of different methods for solving imbalanced problems, in almost all cases, the reweighting method performs well. It also shows similar performance to the baselines in most cases, and the performance on the two datasets analyzed in this section never shows a significant drop, making it a clear recommendation for image classification tasks.

Undersampling shows poor overall performance. In the above experimental scenarios, undersampling shows a significant drop in performance compared to the baseline. We use a direct undersampling method to keep the number of samples in the head class consistent with the number of samples in the tail class, so a large number of samples in the head class are lost and the overall performance is degraded. There is never a significant advantage over oversampling. Balanced sampling and oversampling achieve better performance in CIFAR-10 and CIFAR100 respectively. In the CIFAR-10 scene, each category has a larger number than CIFAR-100, and the classification task is relatively simple. Therefore, the performance of balanced sampling once exceeded the baseline. However, it shows poor overall performance on CIFAR-100 with a relatively insufficient sample size. Oversampling has relatively good performance when there are relatively few samples.

In the re-weighting method, the weighting coefficient of the reciprocal square root is more moderate. At the same time, it can be seen from Table 3 that the weighting method of the reciprocal square root is more stable in different datasets,

and it is particularly prominent in the CIFAR-100 data set. Therefore, the re-weighting method has been deeply studied by deep learning in recent years.

C. Evaluation of ImageNet-Tiny dataset

Experimental results on the ImageNet-Tiny dataset confirm the impact of imbalance on classifier performance. Table 4 compares the performance of the five methods on ImageNet-Tiny. For the multi-class F1-score, the imbalance of class samples decreases from 0.5064 to 0.2004 as the imbalance degree increases. The results confirm that among the sampling methods, the oversampling method performs better than the undersampling method. Compared to the baseline, a drop in F1-score is observed due to the extremely unbalanced oversampling. However, this is strong evidence that oversampling still has disadvantages for extremely imbalanced highly complex tasks. The results also confirm that the reweighting method performs better than the sampling method in all cases for the multi-class F1-score. The resampling strategy even outperforms the baseline in extreme imbalances. Furthermore, the test accuracy on the highest imbalanced training set is only about 40 percent of the original ImageNet-Tiny, through these experimental results it is shown that caution should be exercised when any sampling technique is applied to highly complex tasks with extreme imbalances. a broader study of a specific issue.

V. EXPERIMENT

A. Datasets and models

In our study, 2 benchmark datasets were used: CIFAR-10 and CIFAR-100 [32] created long-tailed CIFAR-10 by exponential imbalance in the form of Eq. 1, CIFAR-100. The test set is kept constant. Figure 2 shows the number of training images under different parameters δ .

All data sets are subjected to the same number of iterations. ResNet-32 was used as our experimental network. All experiments applied the training method in [20].

B. Long-Tailed CIFAR

We have conducted extensive experiments on the long-tailed CIFAR datasets considering five imbalance factors. It is evident from Table 1. (1) The original Class Balanced loss, whose results are strongly influenced by the hyperparameter β . We cannot determine a suitable hyperparameter β from the beginning, and if an inappropriate parameter is chosen, it may even seriously affect the final results. (2) The hyperparameter β in CIFAR-100 is generally smaller than the hyperparameter β in CIFAR-10. As in Figure 3, an adaptive effect can be achieved by the number of samples calculated by our method. There is no need to perform tedious hyperparameter settings manually.

It can be seen from table 1 that, due to the difference in the number of classes and the number of each sample of CIFAR100 and CIFAR10, their recognition accuracy also has a significant difference. The unbalanced change also causes a significant change in recognition accuracy. With the imbalance

TABLE I

THE FIGURE SHOWS THE ERROR RATE OF DIFFERENT LOSS FUNCTIONS IN THE LONG-TAILED CIFAR. INCLUDING THE RESULT OF THE BEST PARAMETER OF CLASS-BALANCED LOSS AND THE RESULT OF ADAPTIVE CLASS-BALANCED LOSS. (SM STANDS FOR SOFTMAX, SGM STANDS FOR SIGMOID). THE ADAPTIVE CLASS BALANCE LOSS SHOWS GOOD EXPERIMENTAL RESULTS, AND IT IS FULLY AHEAD OF THE CLASS BALANCE LOSS IN THE CIFAR10 DATASET.

Dataset Name	Long-Tailed CIFAR-10					Long-Tailed CIFAR-100				
δ	0.005	0.01	0.02	0.05	0.1	0.005	0.01	0.02	0.05	0.1
Softmax	34.32	29.64	25.19	17.77	13.61	65.16	61.68	56.15	48.86	44.29
Sigmoid	34.51	29.55	23.84	16.40	12.97	64.39	61.22	55.85	48.57	44.73
Focal($\gamma=0.5$)	36.00	29.77	23.28	17.11	6.75	65.00	61.31	55.88	48.90	44.30
Focal($\gamma=1.0$)	34.71	29.62	23.29	17.24	13.34	64.38	61.59	55.68	48.05	44.22
Focal($\gamma=2.0$)	35.12	30.41	23.48	16.77	13.68	65.25	61.61	56.30	48.98	45.00
Class-Balanced	31.11	25.43	20.73	15.64	12.51	63.77	60.40	54.68	47.41	42.01
Loss Type	SM	Focal	Focal	SM	SGM	Focal	Focal	SGM	Focal	Focal
β	0.9999	0.9999	0.9999	0.9999	0.9999	0.9	0.9	0.99	0.99	0.999
γ	-	1.0	2.0	-	-	1.0	1.0	-	0.5	0.5
Adaptive Class-Balanced	30.55	25.11	20.56	15.00	12.33	64.76	60.00	54.40	47.68	41.17
Loss Type	SGM	SGM	Focal	Focal	Focal	SM	Focal	Focal	SGM	Focal
γ	-	-	2.0	1.0	0.5	-	1.0	2.0	-	0.5

TABLE II

THE F1-SCORE ACCURACY OF CIFAR DATASET ON RESNET18.

Dataset	Baseline	UnderSampler	OverSampler	BalanceSampler	Reweight-inv	Reweighting-inv _{sq} r
CIFAR10	0.8433	-	-	-	-	-
CIFAR10-0.1	0.7134	0.5577	0.6997	0.7160	0.7138	0.7152
CIFAR10-0.05	0.6388	0.4756	0.6494	0.6631	0.6453	0.6495
CIFAR10-0.02	0.5558	0.3989	0.5435	0.5574	0.4585	0.5558
CIFAR100	0.5607	-	-	-	-	-
CIFAR100-0.1	0.3812	0.2238	0.3523	0.3479	0.3761	0.3760
CIFAR100-0.05	0.3247	0.1546	0.2907	0.2773	0.3149	0.3175
CIFAR100-0.02	0.2579	0.0929	0.2237	0.2150	0.2521	0.2558

TABLE III

THE F1-SCORE ACCURACY OF IMAGENET-TINY DATASET ON RESNET18.

Dataset	Baseline	UnderSampler	OverSampler	BalanceSampler	Reweight-inv	Reweighting-inv _{sq} r
ImageNet-Tiny	0.5064	-	-	-	-	-
ImageNet-Tiny-0.1	0.3021	0.1662	0.2745	0.2647	0.3002	0.2976
ImageNet-Tiny-0.05	0.2532	0.0963	0.2133	0.2046	0.2565	0.2532
ImageNet-Tiny-0.02	0.2004	0.0423	0.1495	0.1393	0.2027	0.2028

from 0.005 to 0.1 in the two data sets, the error rate decreased significantly. The CIFAR100 and CIFAR10 datasets also have significant differences in the number of classes and the number of class samples. Under the same imbalance degree, CIFAR10 has obvious accuracy advantages, which can prove that the number of classes and the number of samples are the key factors that affect the accuracy of classification. In the setting of the parameter β , we can find that when the sample size is sufficient, β tends to 1; that is, the effective sample size keeps the original sample number as much as possible. On the contrary, when the sample size of each class in cifar100 is not so sufficient, the setting of β is more inclined to reduce the effective sample size of the head class.

Our adaptive method outperforms the original weighting method with an effective sample size on the CIFAR10 dataset with sufficient samples and small classification classes. As the number of classes rises and the number of class samples decreases, our method is essentially the same as its original one.

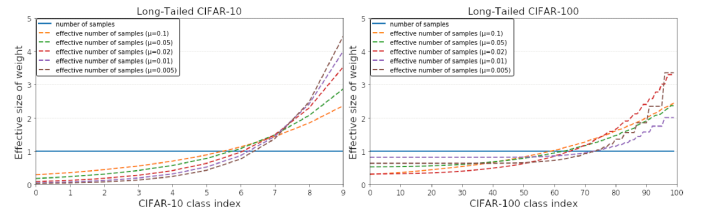


Fig. 4. CIFAT10 and CIFAR100 for different β with an imbalance factor of 0.02. The number of effective samples that are close to the optimal β for different data is calculated by our method.

In Figure 4, we visualize the weighting coefficients to find that as the sample size decreases, we gradually reduce the penalty on the head class and the enhancement on the tail class. This can indicate an over-suppression of the head class or an over-enhancement of the tail class in the case of sufficient samples. It may lead to underfitting of the head class or overfitting of the tail class, which may affect the overall classification

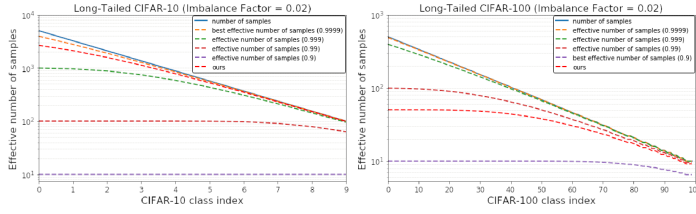


Fig. 5. The magnitude of the weights of each class under different imbalance factors.

effect. Therefore, the re-weighting method should be softer in the case of insufficient class samples than in the case of sufficient class samples.

C. CIFAR

We have conducted experiments on CIFAR10, CIFAR20, and CIFAR100, considering the number of classes and the number of samples, where CIFAR20 is divided by CIFAR100 according to its 20 superclasses. From Table 2, it can be seen that as the number of classes grows and the number of samples decreases, precision drops significantly.

TABLE IV
WITHOUT ADDITIONAL DATA ENHANCEMENT, THE ERROR RATE OF
CIFAR10, CIFAR20, CIFAR100 ON RESNET34.

Number of Samples	500	1000	2500	5000
CIFAR10	75.15	83.29	88.48	92.59
CIFAR20	73.34	80.1	84.07	-
CIFAR100	70.93	-	-	-

VI. DISCUSSION

In this work, Our experimental analysis of the common solutions for sample imbalance in deep learning image classification shows that the reweighting method has some advantages in dealing with the sample imbalance problem. On this basis, we investigate the reweighting method. we consider the crucial factors that affect classification accuracy: the number of samples, the number of classes, and the degree of data imbalance. Most existing studies on the imbalance problem have been concerned with only one or two of these factors. Compared with directly taking the reciprocal of the number of samples or taking the reciprocal of the number of samples, re-weighting is performed, the effective sample method is softer. It can better apply different weights to the head and tail classes. More importantly, we avoid the complicated adjustment parameter problem in the traditional re-weighting method

With long-tail data, the fundamental attributes of the dataset are more prominent for the classification results. The data imbalance is more significant than the general case, because the head class possesses most of the samples in the dataset, while the tail class has a small sample size. As the number of classes increases, the problem worsens. The weights are made to fit the original attributes of the data set better by adaptive methods. The adjustment of the weights by the original attributes may be a new guiding direction for re-weighting methods.

VII. CONCLUSION

This work optimizes the original effective sample calculation method so that hyperparameter settings do not restrict it. The key idea is to incorporate factors that were not considered in the previous work into the weighting considerations to help quantify the number of effective samples more accurately. We conduct extensive experiments on the artificially induced long-tail CIFAR datasets, and the experimental results prove the feasibility and effectiveness of the optimization method.

Our work solves those problems that are not considered in the previous re-weighting work. Furthermore, our improved method does not make any assumptions about the data distribution and can be widely combined with existing loss functions. We hope our work can guide the subsequent re-weighting work.

ACKNOWLEDGMENT

This work was supported in part by the China Chongqing Science and Technology Commission under Grant cstc2020jcsxmsxmX0086, cstc2019jcsx-zdztzx0043, cstc2019jcyj-msxmX0442. China Chongqing Banan District Science and Technology Commission project under Grant 2020QC413, and China Chongqing Municipal Education Commission under Grant KJQN202001137.

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